

ETFT (AutoScientist) — Strategic Presentation Deck

8-12 Slide Narrative · Landscape Format

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SLIDE 1 — The Eroom's Law Problem

Headline: *Science is getting exponentially more expensive — and slower.*

Key visual: Eroom's Law chart — R&D cost per new drug doubles every nine years (mirror image of Moore's Law). Cross-reference: AI research paper volume doubles every 18 months, yet model architectures have converged. More compute, more papers, fewer paradigm shifts.

Speaker note / body copy:

The pharma industry spends \$2.6 billion per approved drug. The ML community produces 400,000 papers per year but ships three genuinely new architectures per decade. The bottleneck is not compute — it is the human cognitive loop that decides *what to experiment on next*. Eroom's Law is not an industry anomaly. It is the default trajectory of any science driven by human intuition.

Call to action: *What if the scientific method itself could be automated?*

SLIDE 2 — The \$100B Blindspot in Every LLM

Headline: *Your AI doesn't know what it doesn't know.*

Key visual: Split diagram — left: LLM frozen weight space (static, dated); right: arXiv preprints published after training cutoff (invisible to model).

Body copy:

Every frontier model — GPT, Claude, Gemini — was trained on data with a hard cutoff. The instant training ends, the model's knowledge begins to decay relative to the frontier.

This is the **Knowledge Blindspot**: the gap between what a model believes is SOTA and what actually is SOTA at inference time.

Current workarounds (RAG, fine-tuning) are **reactive patches**, not architectural solutions. ETFT eliminates the blindspot structurally.

SLIDE 3 — Introducing the Discrete Gradient

Headline: *Algorithms don't appear. They evolve.*

Key visual: Evolutionary tree — horizontal axis: time; vertical axis: fitness $F(a)$. Nodes: a_0 (MLP) $\rightarrow a_1$ (1D CNN) $\rightarrow a_2$ (Depthwise CNN) $\rightarrow a_3$ (MobileNetV2) $\rightarrow a_4$ (SOTA+1 ?)

Each transition labelled with the architectural delta (what changed) and the performance jump (what it bought).

Body copy:

ETFT formalises this intuition mathematically.

Define a fitness function $F(a)$ inversely proportional to validation loss under hard hardware constraints. An **Evolutionary Trajectory** $T = (a_0, a_1, \dots, a_n)$ is an ordered sequence where $F(a_{i-1}) < F(a_i)$.

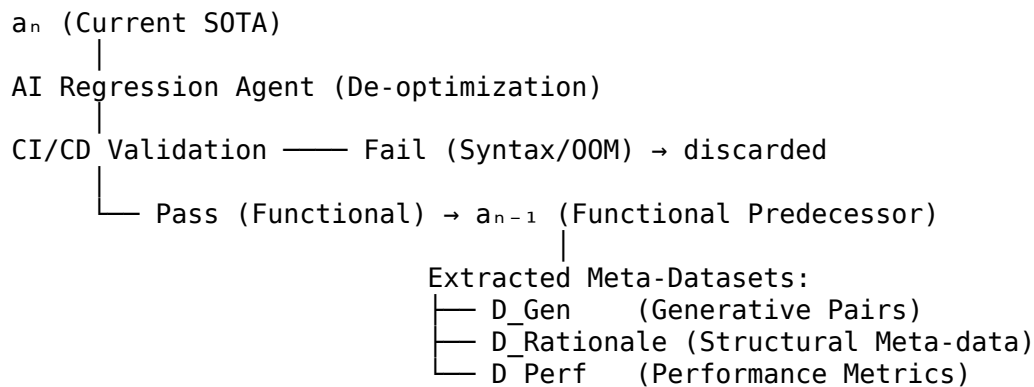
Each transition $a_{i-1} \rightarrow a_i$ is a **macro-evolutionary jump** — a discrete gradient step in architectural space. ETFT trains a model on these regressions, teaching it *the mechanics of invention*, not just the outcomes.

The model learns not what SOTA is. It learns *why* SOTA becomes SOTA.

SLIDE 4 — The Regression Pipeline (Figure 1)

Headline: *Scalable training data — generated autonomously, validated rigorously.*

Key visual: Reproduction of Figure 1 from ETFT whitepaper.



Body copy:

The training corpus is generated by an **AI Regression Agent** that systematically de-optimises SOTA algorithms into their functional predecessors — and records every architectural decision along the way.

CI/CD validation ensures every generated predecessor is compilable and runnable. The result: a self-growing, high-fidelity dataset of the evolutionary history of algorithms — produced at scale, without human labelling.

D_Perf trains the **Probabilistic Heuristic Filter** — an in-silico triage that screens out OOM-risk candidates before expensive GPU runs. Crucially, *failed* candidates continuously enrich the negative training set, making the filter progressively sharper.

SLIDE 5 — Pareto Delta Analysis: Finding the 20%

Headline: *Not all code is created equal. ETFT finds the 20% that drives 80% of progress.*

Key visual: Pareto bar chart — horizontal: architectural components (attention head, normalization layer, activation function, quantization scheme...); vertical: contribution to performance delta ($a_{n-1} \rightarrow a_n$). 80/20 cut-off line drawn.

Body copy:

Rather than real-time computational profiling (expensive, error-prone), ETFT performs a **deterministic retrospective analysis** of historical deltas.

By comparing a_{n-1} vs a_n across the entire evolutionary history, the system quantifies: - Which sub-components were modified in winning transitions? - What magnitude of change drove the largest performance jumps?

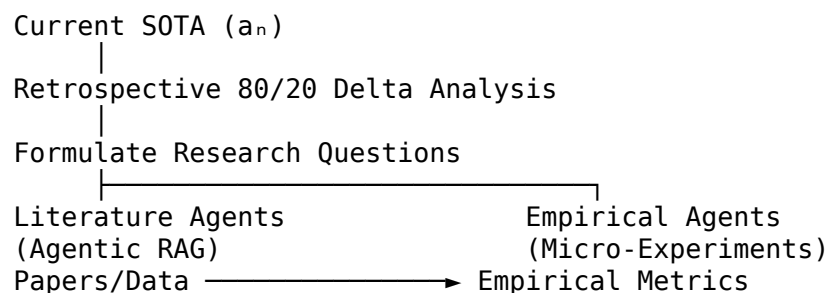
The model learns to recognise the structural signatures of high-yield innovation — then projects that learned 80/20 lens onto the forward trajectory $a_n \rightarrow a_{n+1}$.

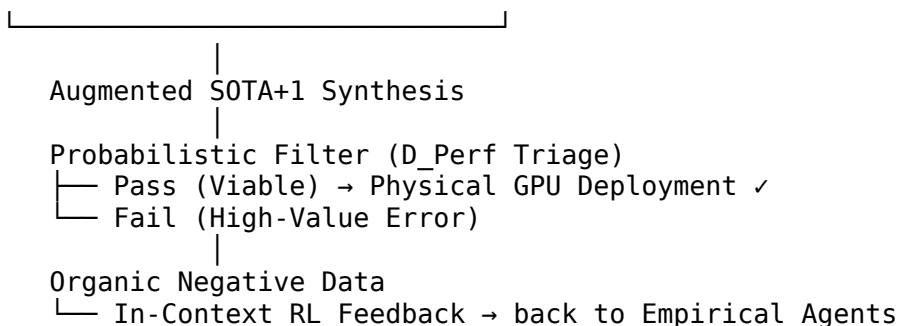
This is not heuristic. It is a **deterministic, data-driven bottleneck oracle**.

SLIDE 6 — The Open-Loop Architecture (Figure 2)

Headline: *A continuously self-improving R&D pipeline — running 24/7.*

Key visual: Reproduction of Figure 2 from ETFT whitepaper — the full Open-Loop Empirical Ecosystem diagram with highlighted feedback loops.





Body copy:

Once the 80/20 bottleneck is identified, two autonomous agent classes deploy in parallel:

1. **Literature Agents (Agentic RAG):** Scour arXiv and technical databases in real time — retrieving knowledge published *after* the model's training cutoff. Parametric blindspot eliminated.
2. **Empirical Agents:** Execute micro-experiments. Physical metrics flow back as RL reward signals. Failed experiments are *not discarded* — they become the highest-quality negative training examples the system can produce.

The loop is self-hardening. Every failure makes the next hypothesis smarter.



SLIDE 7 — Competitive Landscape

Headline: *ETFT vs. the field — a structural difference, not a feature difference.*

Capability	Sakana AI	AutoScience (DeepMind)	ETFT (AutoScientist)
Autonomous hypothesis generation	✓	✓	✓
Knowledge cutoff bypass (live RAG)	✗	Partial	✓ Structural
Recursive negative data flywheel	✗	✗	✓ Native
Probabilistic OOM triage (D_Perf)	✗	✗	✓
Domain-agnostic (beyond DL code)	✗	✗	✓ By design
GPU-Poor validated PoC	✗	✗	✓ (April 2026)
Proven Pareto bottleneck oracle	✗	✗	✓

Key differentiator:

Sakana AI and AutoScience automate *search in hypothesis space*. ETFT automates the **learning of how to search** — a second-order advantage that compounds with every experiment run.

SLIDE 8 — GPU-Poor PoC: Proof It Works Now

Headline: *SOTA+1 in TinyML — autonomously discovered in minutes, on a \$1,500 GPU.*

Key visual: Before/after bar chart.

Metric	Human-designed SOTA (a ₃)	ETFT SOTA+1 (a ₄)
Accuracy	93.0%	94.2%
Peak RAM	18 KB	11 KB
Discovery method	Human researcher, months	Autonomous agent, minutes
Compute required	H100 cluster	Consumer RTX 4090

Body copy:

The “GPU-Poor” PoC deliberately avoids large-scale compute to prove the *cognitive architecture* works independently of hardware:

- Evolutionary trajectory (MLP → MobileNetV2) injected into context
- Agent issued Pareto analysis, literature retrieval, and synthesis directives
- By iteration 3 the agent stabilised a **1D State Space Model (Mamba-style)** discovered from post-cutoff literature — surpassing every human-designed baseline

The PoC is reproducible on a free Google Colab instance.

SLIDE 9 — From Deep Learning to General Innovation Accelerator

Headline: *The same engine that invents neural architectures can evolve anything.*

Key visual: Hierarchy pyramid — three tiers: - Tier 1 (base): Code & algorithm optimisation (validated PoC) - Tier 2 (middle): Scientific blueprints & foundational papers (Second-Order Innovation) - Tier 3 (apex): Organisational structures, societal systems (Meta-Evolution)

Body copy:

An algorithm is any ordered sequence of instructions designed to solve a problem. A pharmaceutical synthesis protocol is an algorithm. A clinical trial design is an algorithm. A global supply chain is an algorithm. An organisation is a macro-algorithm.

Once ETFT can map evolutionary trajectories in one domain, the framework transfers. The **General Innovation Accelerator** does not compete with domain experts — it gives them the autonomous research scientist they never had.

Expanding the objective function to encode human wellbeing as the reward signal opens targets no purely capital-efficiency model can reach.

SLIDE 10 — The Ask & Roadmap

Headline: *What we need. What you get.*

Milestone roadmap:

Phase	Timeline	Deliverable
PoC validated	[DONE] April 2026	TinyML SOTA+1 discovery, reproducible
Seed round	Q2 2026	Compute infra, 3-person core team
Full DL benchmark	Q3 2026	ETFT applied to ImageNet / LLM architectures
Domain expansion	Q4 2026	Pharma / materials science pilot partner
Series A readiness	Q1 2027	Revenue pipeline from IP licensing & API access

The ask: Seed investment to fund 12 months of compute and engineering.

What investors get: - Equity in the IP layer of autonomous scientific discovery - First access to licensing deals with pharma, semiconductor, and defence R&D organisations - A front-row seat to the next Eroom's Law reversal

Links: [Demo video — ETFT_AI_Research_Scientist.mp4] · [Full Technical Whitepaper — available under NDA]

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